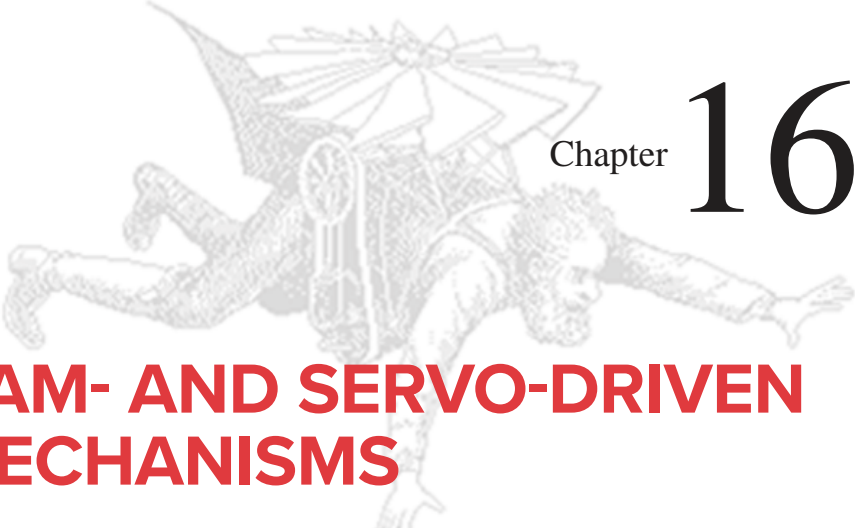




CAM- AND SERVO-DRIVEN MECHANISMS

A professor must have a theory just as a dog must have fleas

H. L. MENCKEN

16.0 INTRODUCTION

A servomechanism uses feedback to control the system's output, such as its position and velocity. A desired control signal is applied to the motor and a transducer measures the output and feeds back information on its actual position. Any difference between the control signal and the fed-back output is an error that is amplified and used to force the system in a direction to reduce or eliminate the error. This is an application of control theory. An example is the cruise-control system in an automobile which takes as input a set speed and feeds back the vehicle's actual speed. Any difference is used to adjust throttle position until the error is brought as close to zero as possible.

Servomotors were introduced briefly in Chapter 2. They now are being used frequently in modern machinery, in part because they have become less costly than in the past. They offer many advantages over conventional motors because they can provide constant speed against dynamic variations in load torque due to their closed-loop operation. An encoder is built into or attached to the motor that provides a large number of equispaced pulses per revolution. This train of pulses is fed back to the computer-driven motor controller, which varies the current to the motor to maintain constant speed or provide any programmed variation in speed. The motor can also be made to hold an angular position against a load, thus creating a dwell on the output motion of any mechanism

driven by the motor. Many motion functions can be programmed into the controller to accelerate the motor to a set speed and decelerate it to a different speed or to zero. It can be made to follow any motion function, similar to a cam-driven follower.

Cams were discussed in Chapter 8 and are used in all kinds of machinery, most often to drive a linkage mechanism. A **function generator** was defined in Chapter 3 as *the correlation of an output motion with an input motion in a mechanism*. The cam-follower mechanism was also noted to be a *flexible and useful function generator*. A **servomotor** can generate any output motion that can be programmed into it and that can be dynamically achieved within the limits of the motor's torque capability. So, we can now add the servomotor to linkage and cam-follower mechanisms as a potential means for function generation. More importantly, the combination of a servomotor and a linkage can create an even more useful device for the solution of some motion generation problems. In fact, in many cases cam-driven-linkage mechanisms are being replaced by servomotor-driven linkages in machinery.

16.1 SERVMOTORS

Both electric and hydraulic servomotors are widely available. We will limit this discussion to electric servomotors, both rotary and linear. The first electric servomotors were direct current (DC) and used an analog tachometer to provide velocity feedback. This allows the motor to run at close to constant speed in the face of dynamically varying torque loads. As the motor slows down or speeds up under changing load, the tachometer signal that is being fed back to the controller provides a measurement of the actual velocity. The difference between the command or setpoint velocity and the actual velocity constitutes an error. The current to the motor is automatically and continuously adjusted to minimize the error between the setpoint velocity and actual velocity.

Modern servomotors come in many varieties, among them DC servomotors based on conventional DC motor designs, alternating current (AC) servomotors based on induction motors, and the so-called brushless servomotor, which uses a synchronous AC motor that locks onto the frequency of the AC supplied. For this, the mains' AC is rectified to DC in the brushless motor's controller then inverted back to AC at different frequencies as dictated by feedback from the digital encoder on the motor. The frequency of the amplified current sent to the motor is varied by the digital motor controller and amplified to control motor speed. Because the motor's angular position is fed back to the motor controller, this type of servomotor can be made to move to a specified angular location, hold that position, and return to zero, all with controlled velocity and acceleration. The train of pulses from the position encoder or resolver is differentiated to obtain information on motor angular velocity in addition to angular position information. The velocity feedback controls motor speed. A block diagram of this system is shown in Figure 16-1.

Linear servomotors are essentially a rotary motor that has been "unwrapped" into a straight line. They are made as both moving coil and moving magnet types. A moving coil linear motor has a coil moving within an assembly of powerful magnets arranged to form a "track." There is an air gap between coil and magnets. An applied DC voltage causes the coil to move with constant force. A moving magnet type swaps the roles of coil and magnets with a magnet being pulled through the coil by its magnetic field. With either type, reversing polarity reverses motion and position feedback is provided by a

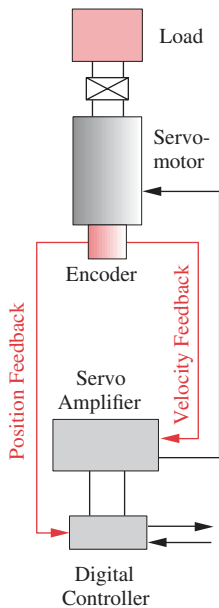


FIGURE 16-1

Typical servomotor system

linear optical or inductive encoder along the track. Linear servomotors, also called linear actuators, are capable of very high force, high speed, and high accelerations. Most linear servomotors require that the moving element be guided in bearings to maintain the air gap between the elements. Modern “Maglev” high speed trains* work on the same principle with the linear motor extending the length of the track and the magnetic flux doing double duty by lifting (levitating) the mass of the train off the track to reduce friction to the very low levels of an air bearing.

16.2 SERVO MOTION CONTROL

We will not address the control of servos beyond the mathematics of their motion control functions. The topic of programming and controlling servos lies well beyond the scope of this text and many books exist that cover this topic in detail. Some are noted in the bibliography of this chapter.

Servo Motion Functions

The same rules surrounding dynamic motion described for cam-follower systems in Chapter 8 also apply to servo motion. Newton’s second law is universal and applies regardless of the means used to provide the motion. Thus the *fundamental law of cam design* defined in Chapter 8 can be restated and applied to servomotor motions as:

The servomechanism function must be piecewise continuous through the first and second derivatives of displacement across the entire interval (360 degrees or one cycle).

Corollary:

The jerk function must be finite across the entire interval (360 degrees or one cycle).

We will call this the *fundamental law of servomechanism design*.

The same set of acceptable functions defined for cams in Chapter 8 can be used for servo motions, and the ones listed as unacceptable in Chapter 8 should be avoided. Unfortunately, some suppliers of servo controllers provide these dynamically inferior functions as choices in their servo programming software. One of the largest suppliers (manufacturer A) offers only two types of acceleration functions in its basic controller, one of which, called trapezoidal velocity, is the same as the constant acceleration (parabolic displacement) function shown in Figure 8-13. This function has infinite jerk at its ends because the acceleration rises instantly from zero to its maximum value and for that reason is rejected as a usable motion function since it violates the fundamental law. The other offering from this supplier is called an “S-curve.”

S-CURVES This term is used by most servo controller manufacturers but is defined differently depending on the manufacturer. The name comes from the fact that any displacement function between two dwells resembles a laid-down letter S as can be seen in Figure 8-23. The same manufacturer A mentioned above defines its S-curve as having constant jerk, linear acceleration, parabolic velocity, and cubic displacement. While this function does not “break the fundamental law,” it is a poor choice dynamically when compared to other possible functions. In fact, their S-curve is actually a trapezoidal acceleration function as shown in Figure 8-14.

* Japan Railways operates *Maglev* (short for Magnetic Levitation) vehicles at speeds up to 350 mph on their 20-kilometer guideway in Yamanashi Prefecture. <http://www.21stcenturysciencetech.com/articles/Summer03/maglev2.html>

TABLE 16-1 Servo Motion Functions Offered by Manufacturer A

Manufacturer's Term	Cam Function Term	Figure	Eqns.	Fund. Law	Comments
Trapezoidal velocity	Constant acceleration	8-13		No	Not an acceptable choice
S-curve	Trapezoidal acceleration	8-14		Yes	Marginally acceptable choice

TABLE 16-2 Some of the Servo Motion Functions Offered by Manufacturer B

Manufacturer's Term	Cam Function Term	Figure	Eqns.	Fund. Law	Comments
Constant position	Dwell			Yes	Match to adjacent segments
Constant velocity	Constant velocity	8-8		No	Match to adjacent segments
Constant acceleration	Constant acceleration	8-13		No	Not an acceptable choice
Simple harmonic	Simple harmonic motion	8-9	8-15a–8-21a	No	Not an acceptable choice
Triple harmonic	Fourier-3 function			Yes	Low harmonic content
Cycloidal (their S-curve)	Cycloidal	8-12	8.7–8.12	Yes	Very smooth, lowest vibration
Modified trapezoidal	Modified trapezoidal	8-15	8-15a–8-21a	Yes	High vibration
Modified sine	Modified sine	8-16	8-15a–8-21a	Yes	Low peak velocity, low vibration
Sine-constant-cosine	SCCA	8-17	8-15a–8-21a	Yes	Depends on b , c , d values
Polynomial	Polynomial	8-25, 8-26	8-24, 8-25	Yes	Depends on boundary conds.
Quadratic spline	Quadratic spline			No	Discontinuous acceleration
Cubic spline	Cubic spline			Yes	Discontinuous jerk

A second major manufacturer B of servo controllers that offers many good choices of functions also has an S-curve but defines it to be sine acceleration or cycloidal displacement, a dynamically superior motion when used between dwells. So the designer needs to be careful when selecting both the controller and the functions offered within that controller's software. Table 16-1 shows the servo functions offered by manufacturer A, and Table 16-2 shows a subset of those offered by manufacturer B. Both tables show the manufacturer's name for the function, the standard cam design name for the same function, and references a figure and equation(s) for that function as defined elsewhere in this text. Each function's compliance with the fundamental law is also noted and comments on the function are provided.

16.3 CAM-DRIVEN LINKAGES

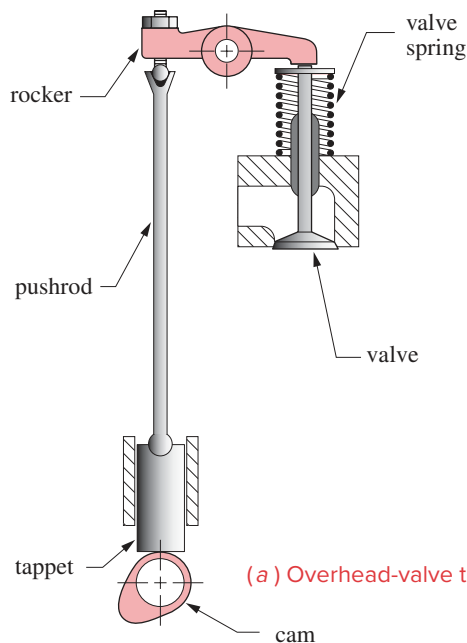
Chapter 8 showed how to design cams to obtain suitable s , v , a , j motions and Chapter 15 showed how to determine dynamic forces and torques in simple cam-follower systems. Both of these chapters were limited to systems using a single translating link as the follower. This is sufficient to explore the basics of cam design, but the fact is that most real cam-follower systems have the cam driving one link of a multibar linkage, referred to as a cam-driven linkage. There are a number of advantages to this arrangement. The cam provides a controlled set of s , v , a , j functions to the input link, and the linkage geometry modifies that motion, perhaps to deliver the motion to a location remote from the cam as shown in Figures 10-11 and 16-2. The linkage can be of any suitable configuration but the most common cam-driven combinations are a fourbar triple-rocker linkage, a fourbar

slider with either the crank or the slider driven by the cam, or a Watt II sixbar with either a rocker or slider output link.

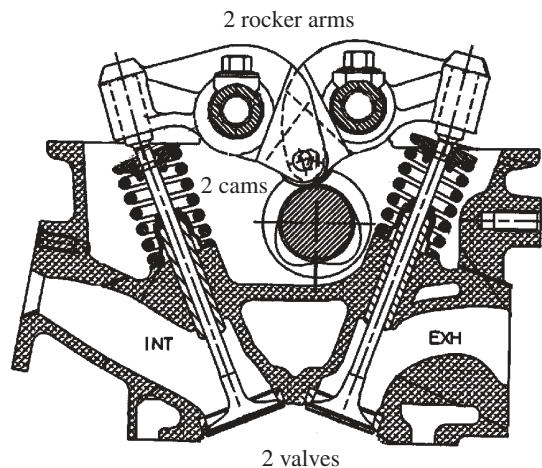
Depending on the particular linkage geometry, it may only slightly modify the s , v , a , j functions as input by the cam, or it could radically alter them. The designer can choose to define the s , v , a , j motion at the cam roller follower and let the linkage geometry modify those functions at the linkage output link (end effector). Or, the designer can choose to apply the desired s , v , a , j motion to the end effector and recalculate the cam shape to account for the linkage's distortion of the motion. In the latter case, the cam will have a different function cut into its shape.

The pushrod overhead valve mechanism of Figure 16-2a is an example of the first type mentioned above, which only slightly modifies the cam function's shape. Even if the rocker has unequal-length arms, the motion of the valve will be essentially the same shape in s , v , a , j as that of the tappet because the tappet motion is parallel to the valve motion and the rocker's midposition is orthogonal to both motions, giving transmission angles of 90° at midstroke. Any ratio in the rocker arms will serve to amplify the magnitudes of s , v , a , j but will essentially preserve their shapes with small distortion.* The overhead camshaft mechanism of Figure 16-2b has a significant geometric change between the cam-roller motion and valve motion due to the differences in angle between the vectors of valve displacement and roller displacement. In this case, the designer applied the desired s , v , a , j functions to the valve-stem end of the rocker and calculated what the corresponding functions needed to be at the cam to get that result given the linkage geometry. The cam was then contoured to generate the modified displacement function that gave the proper kinematic motion at the valve.

* But, remember that any ratio between the cam follower and the end effector will modify the effective mass and spring rate by the square of that ratio.



(a) Overhead-valve train



(b) Overhead-camshaft valve train

FIGURE 16-2

Cam-driven linkages

In either case the task becomes that of combining the kinematics of the chosen s , v , a , j functions, applied at the desired end of the mechanism, with the kinematics of the linkage to obtain the motion of the other end. All the mathematical tools we need for this task have been developed in previous chapters of this book. All that remains is to apply them, which we will now do with some examples.



EXAMPLE 16-1

Cam-Driven Fourbar Slider With Motion Functions Applied to the Input Link

Given:

A cam-driven fourbar crank-slider with the geometry shown in Figure 16-3 is driven by a cam with a constant velocity motion program similar to that developed in Example 8-12 and Figure 8-42. The slider is the end effector and must chase a constant velocity conveyor for at least 6 in at 10 in/sec with minimal velocity error and return to the start position to repeat the cycle.* The motion functions are applied to the cam-follower arm. The linkage was designed in a CAD system, which determined the geometry of Figure 16-3 and that 30° of crank rotation gives 6.435 in of constant velocity motion at 1 Hz.

Problem:

Compare the displacement and velocity functions applied to the cam follower (link 2) and the resulting functions at the end effector of the linkage (link 4).

Solution:

- 1 The equations for the position of the crank-slider linkage were developed in Section 4.6, its velocity in Section 6.7, and acceleration in Section 7.3 and are repeated here for your convenience.

$$\theta_{31} = \arcsin\left(\frac{a \sin \theta_2 - c}{b}\right) \quad (4.16a)$$

$$d = a \cos \theta_2 - b \cos \theta_3 \quad (4.16b)$$

$$\omega_3 = \frac{a \cos \theta_2}{b \cos \theta_3} \omega_2 \quad (6.22a)$$

$$\dot{d} = -a \omega_2 \sin \theta_2 + b \omega_3 \sin \theta_3 \quad (6.22b)$$

$$\alpha_3 = \frac{a \alpha_2 \cos \theta_2 - a \omega_2^2 \sin \theta_2 + b \omega_3^2 \sin \theta_3}{b \cos \theta_3} \quad (7.16d)$$

$$\ddot{d} = -a \alpha_2 \sin \theta_2 - a \omega_2^2 \cos \theta_2 + b \alpha_3 \sin \theta_3 + b \omega_3^2 \cos \theta_3 \quad (7.16e)$$

The terminology used in the equations is shown in Figure 16-3.

- 2 The cam program has two segments: 220° of constant velocity at $49.066 \text{ } 05^\circ/\text{s}$ for 30° of follower arm (crank) motion, and a fifth-degree polynomial return motion over 140° . Total angular displacement of the crank is 33.2° in segment 2.
- 3 The boundary conditions for the polynomial in segment 1 are shown in Table 16-3. The boundary conditions for the polynomial in segment 2 are shown in Table 16-4. The normalized displacement equations for the cam functions as derived with program DYNACAM with $x = \theta/\beta$ are as follows:[†]

* The reader might reasonably wonder why anyone would bother to add the complexity of a linkage to the system when the same output motion could be obtained from a cam directly driving a translating follower. One reason would be that it was not possible to mount and power the cam adjacent to the end effector. Another might be that the stroke would require a large cam to get acceptable pressure angles. One can multiply the stroke of the cam follower by adding a linkage.

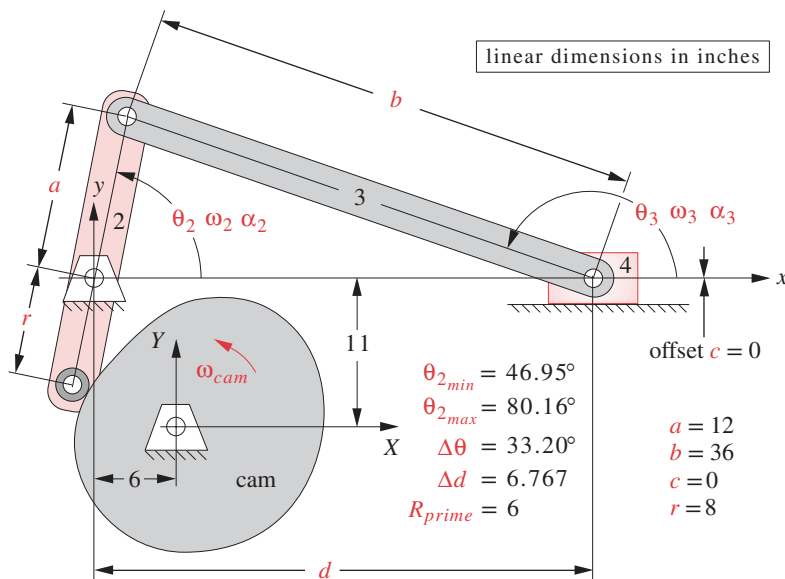


FIGURE 16-3

Cam-driven slider-crank linkage

TABLE 16-3
Boundary
Conditions

Example 16-1, Segment 1

When $\theta = 0$:

$$s = 0^\circ, \\ v = 49.066 \text{ } 05^\circ/\text{sec}$$

TABLE 16-4
Boundary
Conditions

Example 16-1, Segment 2

When $\theta = 0$:

$$s = 30.000 \text{ } 05^\circ, \\ v = 49.066 \text{ } 05^\circ/\text{sec}, \\ a = 0$$

When $\theta = 140^\circ$:

$$s = 0^\circ, \\ v = 49.066 \text{ } 05^\circ/\text{sec} \\ a = 0$$

Segment 1:

$$S = 30.00x \text{ deg}$$

$$V = \frac{\omega}{\beta} (30) = \frac{2\pi}{3.8397} (30) = 49.066 \text{ deg/sec} \quad (16.1)$$

$$A = \frac{\omega^2}{\beta^2} (0) \text{ deg/sec}^2$$

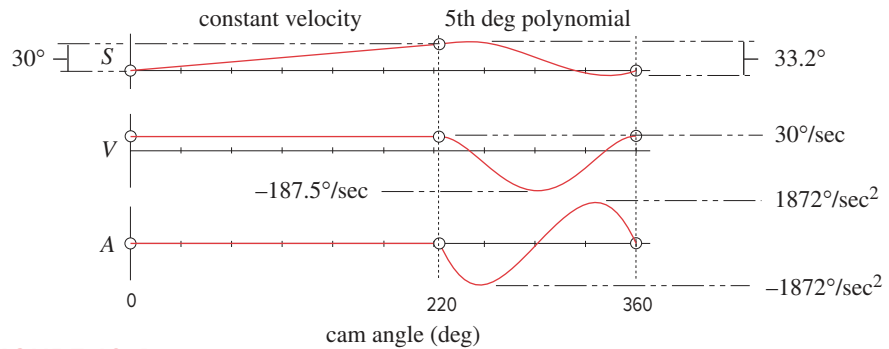
Segment 2:

$$S = -294.546x^5 + 736.364x^4 - 490.909x^3 + 19.091x + 30.00 \text{ deg}$$

$$V = \frac{\omega}{\beta} (-1472.73x^4 + 2495.46x^3 - 1472.73x^2 + 19.091) \text{ deg/sec} \quad (16.2)$$

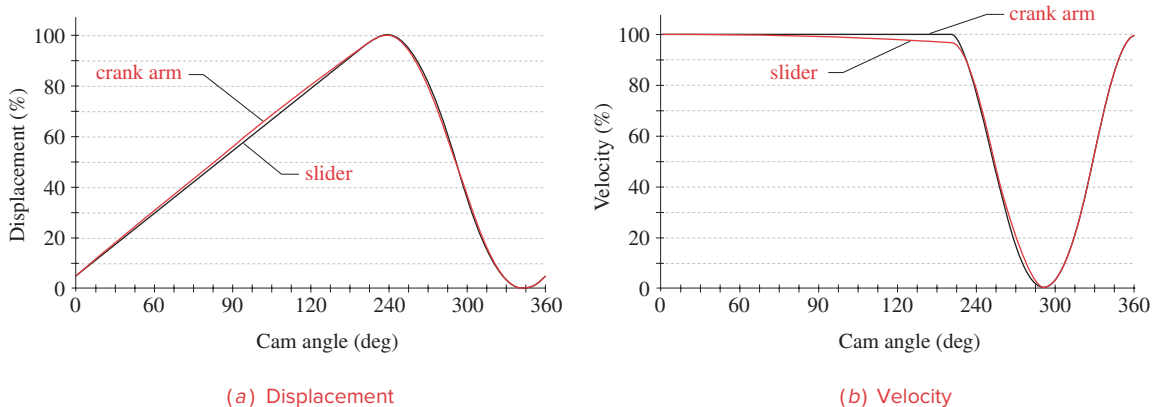
$$A = \frac{\omega^2}{\beta^2} (-5890.91x^3 + 7486.38x^2 - 2945.46x) \text{ deg/sec}^2$$

- 4 These S , V , A equations are the driving functions for the input crank of the linkage where $\omega = 2\pi \text{ rad/sec}$ for a 1-Hz cycle, and β is the angular duration of the motion in radians—3.8397 rad for segment 1 and 2.4435 rad for segment 2.
- 5 The equations for the kinematic behavior of the linkage from step 1 must be solved at a set of discrete points (e.g., every degree of the cam cycle) to determine the behavior of the slider d . Equation 16.1 applies to the first 220° (constant velocity) and equation 16.2 to the last 140° .
- 6 This is obviously a job for a computer and one was used to solve these equations. A good approach is to use an equation solver such as *Matlab*, *Mathcad*, or *TKSolver*. Plots of the polynomial motions from program DYNACAM are shown in Figure 16-4.

**FIGURE 16-4**

S, V, A functions as applied to cam-follower arm in Example 16-1

- 7 Figure 16-5a shows the displacement of the follower arm and the slider superposed and normalized to percent for comparison. The nonlinear behavior of the linkage distorts the entire function. Figure 16-5b shows the normalized velocities of follower arm and slider superposed. Note that the follower arm has constant angular velocity during the first segment but the slider does not have constant linear velocity.
- 8 Figure 16-6 shows the percent error between the crank arm and slider in both displacement and velocity. The differences in this example are not extreme, mainly because the crank only rotates through 30° during the constant velocity motion. The larger the angle of crank rotation, the larger the distortion in the output functions.
- 9 If the magnitude of the deviation from constant velocity is small enough to still allow proper function, then this solution will be adequate as is. If the error is not tolerable, then the solution is to apply the desired motion function to the slider and calculate the linkage's deviation in reverse to redefine the cam function to correct it. The next example will do this.

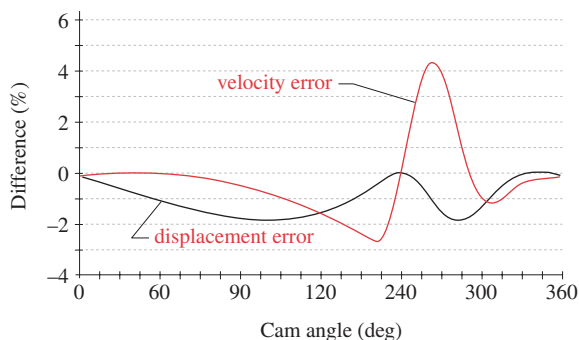


(a) Displacement

(b) Velocity

FIGURE 16-5

Normalized input and output displacement and velocity for Example 16-1

**FIGURE 16-6**

Normalized error in displacement and velocity for Example 16-1

**EXAMPLE 16-2**

Cam-Driven Fourbar Slider With Motion Functions Applied to the Output Link

Given: A cam-driven fourbar crank-slider with the geometry shown in Figure 16-3 is driven by a cam with a constant velocity motion program similar to that developed in Example 8-12 and Figure 8-42. The slider is the end effector and must chase a constant velocity conveyor for at least 6 inches at 10 in/sec with minimal velocity error and then return to the start position to repeat the cycle. The motion functions are applied to the slider and converted to modified functions to be applied to the cam-follower arm with a modified contour cam running at 1 Hz.

Problem: Compare the displacement, velocity, and acceleration functions between the cam follower (link 2) and the end effector of the linkage (link 4) and compare the cam profile for this solution to that of Example 16-1.

Solution:

- 1 The linkage geometry is the same as shown in Figure 16-3.
- 2 The equations that define the motion of the slider-crank when driven at the slider rather than at the crank take on a different form than those shown in Example 16-1. The vector loop equation for both cases is the same, so equations 4.14a, b, c, and 4.15a, b apply to both cases. The derivation starts with the above noted equations for either mode of operation. The simultaneous solution of equations 4.15a, b for the crank-driven mode is done by simple substitution as shown in equations 4.16a, b. However, solving for the slider-driven mode with slider position d as the input is more complex because both variables now sought, θ_2 and θ_3 , are contained in transcendental expressions in both equations 4.15a and 4.14b. This requires the derivation done in equations 4.18 to 4.21 to find θ_2 and θ_3 as functions of slider position d .
- 3 The derivation for crank velocity is done in equations 6.24 and for crank acceleration in equations 7.17. The results of these derivations for θ_2 , ω_2 , and α_2 , as a function of slider position d , velocity $\dot{d}$, acceleration $\ddot{d}$, and linkage geometry a, b, c (shown in Figure 16-3) are used in the following steps.

$$\begin{aligned}
K_1 &= a^2 - b^2 + c^2 + d^2 \\
K_2 &= -2ac \\
K_3 &= -2ad \\
A &= K_1 - K_3 \\
B &= 2K_2 \\
C &= K_1 + K_3 \\
\theta_{2,2} &= 2 \arctan \left(\frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \right)
\end{aligned} \tag{16.3}$$

- 4 The angular velocities ω_2 and ω_3 as a function of slider velocity and link geometry are:

$$\begin{aligned}
\omega_2 &= \frac{\dot{d} \cos \theta_3}{a(\cos \theta_2 \sin \theta_3 - \sin \theta_2 \cos \theta_3)} \\
\omega_3 &= \frac{a \omega_2 \cos \theta_2}{b \cos \theta_3}
\end{aligned} \tag{16.4}$$

- 5 The angular accelerations α_2 and α_3 as a function of slider acceleration and link geometry are:

$$\begin{aligned}
\alpha_2 &= \frac{b \omega_3^2 - a \omega_2^2 \cos(\theta_2 - \theta_3) - \ddot{d} \cos \theta_3}{a \sin(\theta_2 - \theta_3)} \\
\alpha_3 &= \frac{a \alpha_2 \cos \theta_2 - a \omega_2^2 \sin \theta_2 + b \omega_3^2 \sin \theta_3}{b \cos \theta_3}
\end{aligned} \tag{16.5}$$

TABLE 16-5
Boundary
Conditions

Example 16-2, Segment 1

When $\theta = 0$:

$$\begin{aligned}
S &= 0, \\
V &= 6.114 \text{ in/sec}
\end{aligned}$$

TABLE 16-6
Boundary
Conditions

Example 16-2, Segment 2

When $\theta = 0$:

$$\begin{aligned}
s &= 6.114 \text{ 210 in}, \\
v &= 10 \text{ in/sec}, \\
a &= 0
\end{aligned}$$

When $\theta = 140^\circ$:

$$\begin{aligned}
s &= 0, \\
v &= 10 \text{ in/sec}, \\
a &= 0
\end{aligned}$$

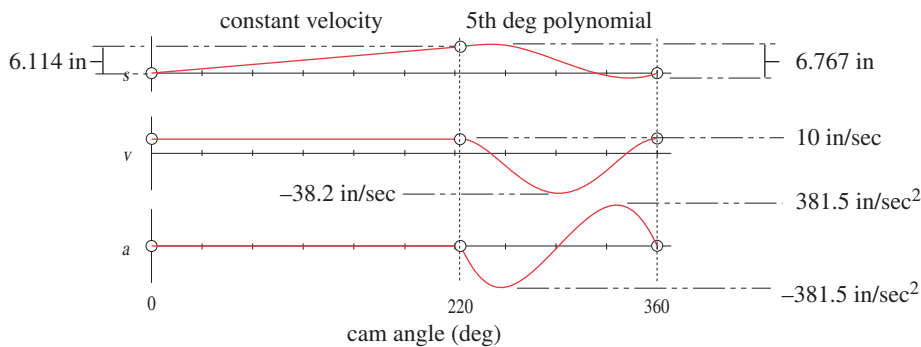
- 6 The cam program has two segments: 220° of constant velocity at 10 in/s over 30° of follower arm (crank) motion, and a fifth-degree polynomial return motion over 140° . Total displacement is 6.767 in.
- 7 The boundary conditions for the polynomial in segment 1 are shown in Table 16-5. The boundary conditions for the polynomial in segment 2 are shown in Table 16-6. The s , v , a functions as derived with program DYNACAM are:

Segment 1:

$$\begin{aligned}
S &= 6.114x \text{ in} \\
V &= \frac{\omega}{\beta}(6.114) = \frac{2\pi}{3.8397}(6.114) = 10 \text{ in/sec} \\
A &= 0
\end{aligned} \tag{16.6}$$

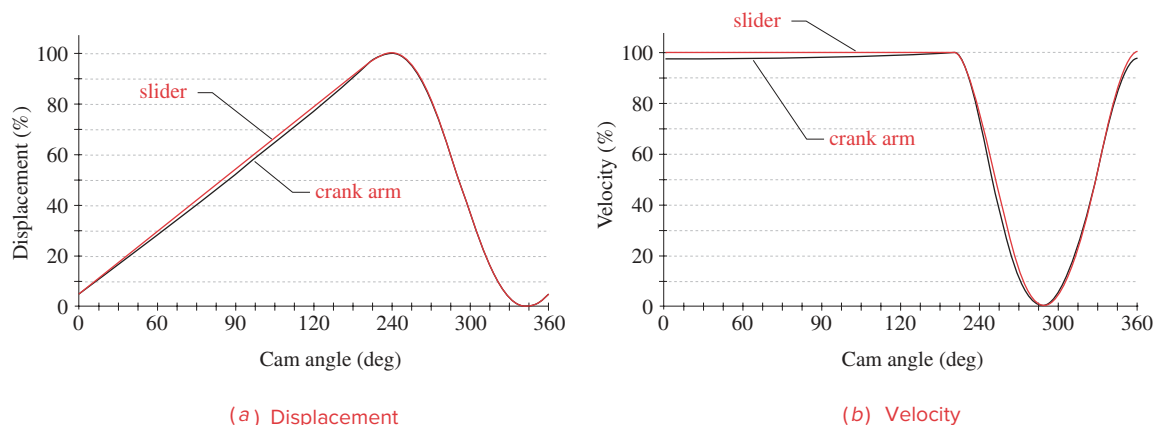
Segment 2:

$$\begin{aligned}
S &= -60.029x^5 + 150.073x^4 - 100.049x^3 + 3.891x + 6.114 \text{ in} \\
V &= \frac{\omega}{\beta}(-300.145x^4 + 600.292x^3 - 300.147x^2 + 3.891) \text{ in/sec} \\
A &= \frac{\omega^2}{\beta^2}(-1200.58x^3 + 1800.88x^2 - 600.294x) \text{ in/sec}^2
\end{aligned} \tag{16.7}$$

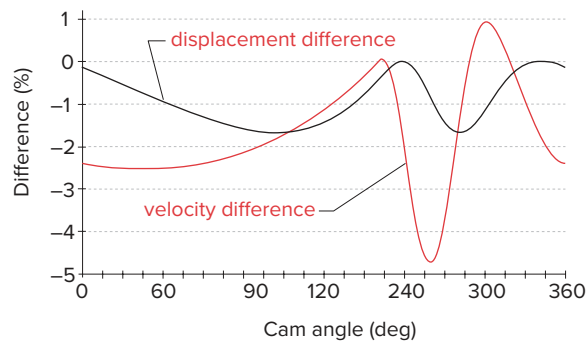
**FIGURE 16-7**

S, V, A functions as applied to slider in Example 16-2

- 8 Equation 16.6 applies to the first 220° and equation 16.7 to the last 140° . The normalized variable $x = \theta/\beta$ runs from 0 to 1. The S, V, A diagrams for these polynomial motions as applied to the slider are shown in Figure 16-7. Note that the displacement is now in length units rather than degrees as was the case in the previous example (Figure 16-4).
- 9 The equations for the kinematic behavior of the linkage from equations 16.3 through 16.5 must be solved at a set of discrete points (e.g., every degree of the cam cycle) using the functions of equations 16.7 and 16.8 as input to the slider to determine the behavior of the crank arm. The crank arm displacement is then used to calculate the required cam profile.
- 10 This is obviously a job for a computer and one was used to solve these equations. A good approach is to use an equation solver such as *Matlab*, *Mathcad*, or *TkSolver*. The problem can be solved numerically using discrete representations of the motion functions calculated at every degree or fraction thereof.
- 11 Figure 16-8a shows the displacement of the slider and the crank arm superposed and normalized to percent for comparison. The nonlinear behavior of the linkage distorts the entire function. Figure 16-8b shows the normalized velocities of follower arm and slider superposed.

**FIGURE 16-8**

Normalized input and output displacement and velocity for Example 16-2

**FIGURE 16-9**

Difference in displacement and velocity between slider and cam follower in Example 16-2

Note that the slider has constant linear velocity during the first segment but the crank arm does not have constant angular velocity.

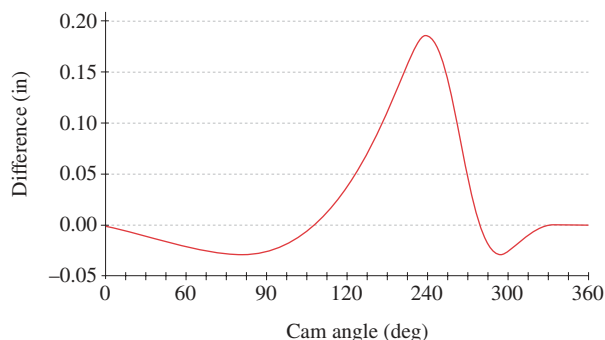
- 12 Figure 16-9 shows the percent difference between the crank arm and slider in both displacement and velocity. The distorted crank arm functions serve to correct the error introduced by linkage geometry and give the designed motions at the slider.
- 13 Figure 16-10 shows the difference in radius to the roller center for the cam of Example 16-1 and that of Example 16-2. The first has the desired functions applied to its follower. The second has the functions applied to the slider with the cam profile recalculated to remove the linkage geometry error. The largest radial difference between the cams is nearly 0.2 in.

Whatever the linkage configuration, the exercise outlined in the above two examples will apply. The cam-motion functions and the equations that define the linkage motion transfer function must be combined, working from input to output, or vice versa, depending on whether you want to apply the theoretical cam functions to the input or output ends of the mechanism.

16.4 SERVO-DRIVEN LINKAGES

Applications that have traditionally used cam-driven linkages for pick-and-place operations, conveyor chasing motions, and other assembly tasks are increasingly using linkage mechanisms directly driven by a servomotor through a gear reducer. The pros and cons of this approach versus the traditional cam-driven linkage are addressed in a later section of this chapter.

If the need is simply to move the end effector of the linkage from a start position (perhaps in a dwell) to a final position and hold it there in dwell before returning, then appropriate mathematical functions from the collection of acceptable double-dwell cam-follower functions can be applied directly to the servo axis. This situation is analogous to the *critical extreme position* (CEP) cam design case described in Chapter 8 where the path motion between the end positions is not critical. Nevertheless, when selecting motion functions to drive the servo axis, be sure to observe the *fundamental law of servo-*

**FIGURE 16-10**

Difference in cam profile radius with functions applied to cam follower versus end effector

mechanism design described in an earlier section of this chapter. The distortion of motion between the endpoints introduced by the linkage may not be detrimental to function in the CEP case except to the degree that it increases accelerations on the members. Links must be designed to withstand dynamic loads from their accelerations and motors selected to provide the needed torque. Most mechanical failures of this type of mechanism occur during acceleration when starting or stopping, especially during an emergency stop when the servomotors must dynamically brake the system to a stop in one or a few machine cycles.

In cases where critical intermediate positions or velocities are specified between the endpoints, distortion of the motion functions can be a problem. This is analogous to the *critical path motion* (CPM) case described for cam mechanisms in Chapter 8. Also, if the input link driven by the servomotor (i.e., the crank of a crank-slider, fourbar, or sixbar mechanism) needs to rotate through a significant angle, it will give greater distortion than was the case in the two previous examples. In such cases, it can become critical to modify the servomotor's control function to account for the linkage geometry's distortion and create the desired precision motion at the linkage end effector. The best way to illustrate this process is with an example.

**EXAMPLE 16-3**

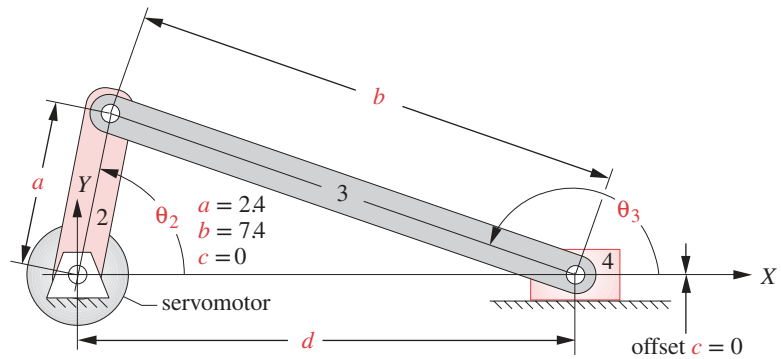
Slider-Crank Linkage Driven by a Servomotor to Perform a Forming Operation

Given:

A slider-crank linkage mechanism (Figure 16-11) has been designed to provide the geometry for a suitable motion of a die in a forming press. The required motion begins from a dwell and moves the slider through a 2-in stroke ending at TDC with crank angle $\theta_2 = 0$. The die needs to be brought to zero velocity for an instant at 1.588-in stroke then advanced into the material. It is then pulled back to dwell at 1.400 in before returning to zero. The slider motion timing diagram is shown in Figure 16-11. Rather than design a custom cam to provide the motion and dwells to the linkage crank, it is desired to use a servomotor directly driving the input link of the crank-slider mechanism through a gearbox to obtain the necessary motions and dwells. Machine speed is 120 cycles per min or 2 Hz.

Find:

The motion function needed to program a servomotor to drive a crank-slider linkage that provides the required linkage output motion program as defined at the slider.

**FIGURE 16-11**

Servomotor-driven slider-crank linkage

TABLE 16-7
Boundary
Conditions

Example 16-3, Segment 1

When $\theta = 0^\circ$:

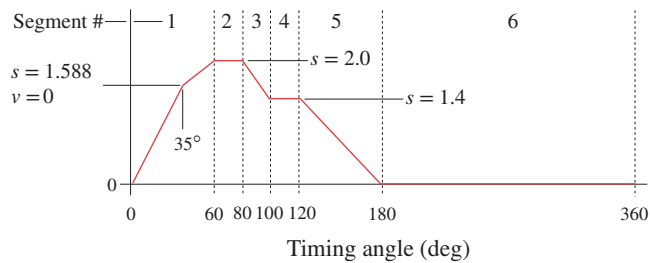
$$s = 0, v = 0, a = 0$$

When $\theta = 35^\circ$:

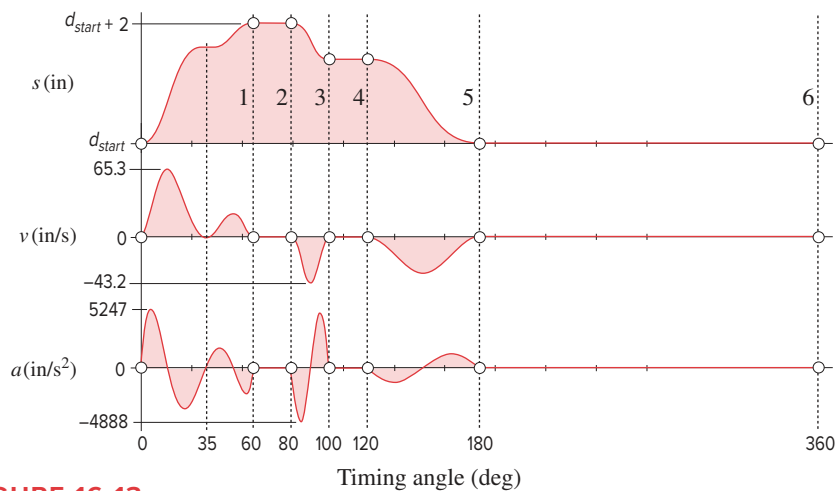
$$s = 1.588, v = 0$$

When $\theta = 60^\circ$:

$$s = 2, v = 0, a = 0$$

**FIGURE 16-12**

Timing diagram for slider in Example 16-3

**FIGURE 16-13** S, V, A functions as applied to slider in Example 16-3

Assume: The position and velocity accuracy required at intermediate points in the stroke requires that the motion function be applied to the slider and mathematically transformed to a function at the servo axis that accounts for linkage geometry.

Solution:

- 1 The functions chosen to provide the motions defined in Figure 16-12 are:

Segment 1: Polynomial over 60° with the boundary conditions of Table 16-7.
 Segment 2: Dwell at 2 in for 20° .
 Segment 3: Cycloidal fall to 1.4 in over 20° .
 Segment 4: Dwell at 1.4 in for 20° .
 Segment 5: Cycloidal fall to zero over 60° .
 Segment 6: Dwell at zero in for 180° .

The S , V , A functions for the slider motion as calculated with program DYNACAM are shown in Figure 16-13.

- 2 The geometry of the slider-crank linkage to be driven by the servomotor was chosen based on packaging requirements and is shown in Figure 16-13. The maximum displacement at TDC when $\theta_2 = 0$ is $d = a + b = 9.8$ in. A 2-in stroke requires that the starting displacement $d_{start} = 9.8 - 2 = 7.8$ in. Equations 16.4 are used to find the starting crank angle at d_{start} .

$$\begin{aligned} K_1 &= a^2 - b^2 + c^2 + d^2 = 2.4^2 - 7.4^2 + 0 + 7.8^2 = 11.840 \\ K_2 &= -2ac = -2(2.4)(0) = 0 \\ K_3 &= -2ad = -2(2.4)(7.8) = -37.440 \\ A &= K_1 - K_3 = 11.840 - (-37.440) = 49.280 \\ B &= 2K_2 = 2(0) = 0 \\ C &= K_1 + K_3 = 11.840 + (-37.440) = -25.600 \end{aligned} \quad (a)$$

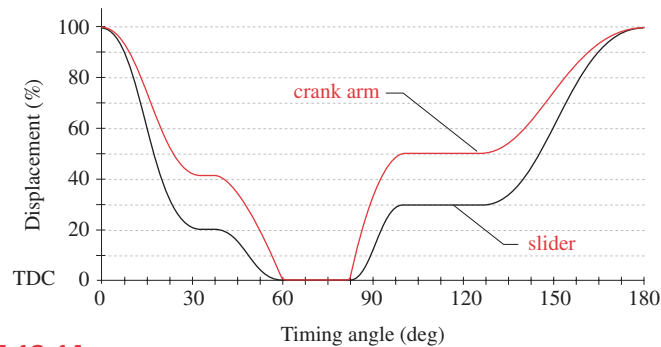
$$\begin{aligned} \theta_{start_{1,2}} &= 2 \arctan \left(\frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \right) \\ &= 2 \arctan \left(\frac{0 \pm \sqrt{0 - 4(49.280)(-25.600)}}{2(49.280)} \right) \\ \theta_{start_1} &= 71.564^\circ \quad \theta_{start_2} = -71.564^\circ \end{aligned} \quad (b)$$

Because of symmetry, either value could be used depending on the desired direction of crank rotation. We will choose clockwise crank rotation with $\theta_{start} = 71.564^\circ$, $\theta_{end} = 0^\circ$.

- 3 It is only necessary to convert the displacement function with the linkage geometry in order to create a usable servo control function because the typical servomotor controller needs only that function for control.* Note that the effects of the chosen velocity, acceleration, and jerk functions are all contained within the displacement function because it comes from the integration of those functions.† Most servo controllers are capable of importing a table of displacement data, which in this case will be generated by the displacement function of Figure 16-12 as modified by the linkage geometry. This data table of crank angles can be generated at any resolution desired from every degree to any fraction of a degree.

* Note that this is the same situation when a cam profile is generated. Only the displacement function is used to create the cam contour, but it contains all the mathematics of the higher derivatives used in the cam design.

† The reason the velocity function was converted in Examples 16-1 and 16-2 was to show the effects of linkage geometry on the constant velocity portion of the functions. Only the displacement function would be used to program the servo controller. The slope of the displacement function encodes the velocity.

**FIGURE 16-14**

Normalized input and output displacements for crank and slider of Example 16-3

- 4 The values of $d_{start} + d$ as defined by the linear displacement function of Figure 16-12 for each increment of machine angle over the cycle from 0 to 360° are used as input to equation 16.4 to calculate their corresponding angular displacements θ_2 to be applied to the servo axis. Do not confuse the linkage crank angles θ_2 with the timing angles shown as the independent variable in Figures 16-12 and 16-14. Those angles represent fractions of one machine cycle, which is always one revolution of the main timing shaft of the machine. In a fully servomotor-controlled machine with no physical timing shaft, this is often an “imaginary” or virtual “master” axis that serves as a timing reference for all “slave” servomotors to follow.
- 5 Figure 16-14 shows the displacement functions for both the slider linear displacement and the linkage crankshaft angular displacement over the first five segments. The 180° dwell of segment 6 is omitted since the crank is then stationary at the start position. Each function is normalized to a range of 0 to 1 to allow their superposition. Note the large differences between the two functions as compared to those of the previous two examples. In those cases, the total rotation of the crank was less and avoided the regions near BDC or TDC where geometric distortion is greatest. This example takes the crank to TDC, and the differences between the functions is seen to increase as it approaches TDC.
- 6 While this design does what the client requested, a close inspection of the crank arm displacement function of Figure 16-14 at 60 and 80 degrees of timing angle shows this to be a dynamically poor solution. There is such a rapid change of slope at those points that it will have very large acceleration. It comes close to violating the fundamental law described earlier in this chapter. This results from the large amount of distortion introduced by the nonlinear slider-crank function. An improvement will be sought in the next example.



EXAMPLE 16-4

Improvement to the Design of Example 16-3

Given:

The design of Example 16-3 proved to be dynamically poor. In that example the required position and velocity constraints were applied to the slider motion and the resulting function was transformed by equations relating slider motion to crank motion. The constraints require that the slider move through a 2-in stroke ending at

TDC with crank angle $\theta_2 = 0$. The slider needs to be brought to zero velocity for an instant at 1.588-in stroke then advanced into the material. It is then pulled back to dwell at 1.400 in before returning to zero. Rather than apply those constraints directly to the slider motion as was done in Example 16-3, we will instead transform the constraints from linear parameters to angular parameters first and then apply those angular parameters to the motion of the servo-driven crank arm. Machine speed is 120 cycles per min or 2 Hz.

Find: The motion function needed to program a servomotor to drive a crank-slider linkage that provides the required linkage output motion parameters as defined for specific positions of the slider.

Assume: The position and velocity accuracy required at intermediate points in the stroke only requires that those specific parameters be transformed to corresponding parameters at the servo axis. The shape of the functions between those points is not critical.

- 1 The functions chosen to provide the motions will be similar to those defined in step 1 of Example 16-3 but will instead be applied directly to the crank arm motion.
- 2 The start and end angles $\theta_{start} = 71.564^\circ$, $\theta_{end} = 0^\circ$ required to achieve the 2-in stroke were calculated in Step 2 of Example 16-3 and will be the same here.
- 3 The crank angles corresponding to the intermediate positions can be calculated by the same method used for the start angle in step 2 of Example 16-3. There must be a point d_1 having instantaneous zero velocity when the slider has moved through 1.588 in of its 2-in stroke. The start position of the slider as referenced to the crank pivot was calculated in step 2 of Example 16-3 to be 7.8 in. The point d_1 will be at $7.8 + 1.588 = 9.388$ in. The corresponding crank angle for that position is found from equations 16.4 to be 29.721° .
- 4 The dwell at a slider stroke of 1.4 in is $7.8 + 1.4 = 9.6$ in from the crank pivot. Equations 16.4 show this slider position to correspond to a crank angle of 36.10° .
- 5 The boundary conditions for crank motion then become:
 - Start at crank angle of 71.564° .
 - Segment 1: B-spline over 60° with the boundary conditions of Table 16-8.*
 - Segment 2: Dwell at zero crank degrees for 20° .
 - Segment 3: Cycloidal rise to 36.19° over 20° .
 - Segment 4: Dwell at 36.19° for 20° .
 - Segment 5: Cycloidal rise to 71.564° over 60° .
 - Segment 6: Dwell at 71.564° for 180° .
- 6 The resulting function superposed on the solution for Example 16-3 is shown in Figure 16-15. Note the smoother contour approaching 60 and leaving 80 timing degrees. The slope discontinuity is gone but the critical points are still being maintained. This is a dynamically superior design to that of Example 16-3.
- 7 It should now be apparent that if any intermediate positions or velocities during the stroke are critical, it is necessary to modify the servo function by the linkage geometry to ensure proper function. If only the end positions are critical (the CEP case), then one may be able to get away with applying the chosen motion functions directly to the input crank. This could nevertheless cause unwanted distortion to the acceleration function and negatively affect the torque function as well. The engineer would be wise to check these effects in each case.

TABLE 16-8
Boundary
Conditions

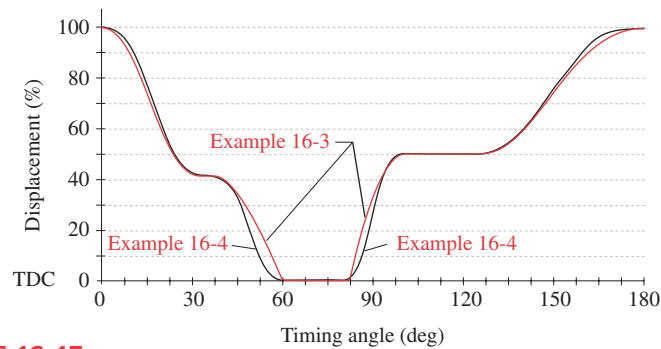
Example 16-4, Segment 1

When $\theta = 0$:
 $s = 71.546^\circ$, $v = 0$, $a = 0$

When $\theta = 35^\circ$:
 $s = 29.721^\circ$, $v = 0$

When $\theta = 60^\circ$:
 $s = 0^\circ$, $v = 0$, $a = 0$

* This set of boundary conditions did not allow an acceptable polynomial solution, and a B-spline was needed to solve it. B-splines give better control over motions and their derivatives than polynomials. They are not described in this book, but information on their mathematics and use can be found in reference 2.

**FIGURE 16-15**

Normalized input displacements for the cranks of Examples 16-3 and 16-4

16.5 OTHER LINKAGES

The approach used for a servo-driven slider-crank linkage in the previous example also can be used for any other linkage mechanism. For a fourbar linkage, the desired motion functions will be applied to the output rocker, link 4, and modified by the linkage geometry as defined by equations 4.8a, 4.10a, and 4.10b. Note that the variables θ_2 and θ_4 must be temporarily interchanged since we want to solve for θ_2 as a function of θ_4 to which the servo motion is applied. Simply consider link 4 to be link 2 and vice versa for the purpose of this calculation. A Watt II sixbar can be solved in the same manner, considering the output link to which the servo functions are to be applied as the temporary “input” link in order to modify the applied servo function to one that can be applied at the true input crank of the mechanism to generate the proper motion.

16.6 CAM-DRIVEN VERSUS SERVO-DRIVEN MECHANISMS

One can sometimes hear the comment among engineers “Cams are obsolete. I can do anything with a servo that you can with a cam.” The author heard this said 40 years ago and occasionally still does today. Is this really true? If so, why are cams still used? Cams have been around a long time. Leonardo DaVinci’s sketches show many examples of cams. Even older illustrations from ancient civilizations show cam-like devices used for various purposes. Servomechanisms are more recent, developed in the second half of the twentieth century. Originally they were only rotary devices (motors) but now are also available as linear actuators. Both systems are extremely useful and valuable, but each has its strong points and limitations.

Flexibility

The greatest strength of servo systems is their inherent programmability. With moderate effort by skilled personnel, a servo can be reprogrammed in a short time to provide a different motion within the limits of its mechanical package. The position, velocity, and acceleration profiles of its motion can be changed in software and more than one version can be stored in memory, allowing the operator to change the motion with a button press.

A cam can provide the same motions as a servo, but its motion program is encoded “in hardware” in its shape. To change the motion program requires remachining its contour. Since its motion program and contour are typically designed with software, and modern manufacturing techniques allow a cam to be made in a day or two, the cam motion is capable of change as well. But it requires new hardware.

Cost

Servomotors and their controllers are expensive, typically costing anywhere from \$1000 for a 1/4 hp example to thousands of dollars for large examples. Cams can cost anywhere from a few hundred dollars for an open radial cam to several times as much for a more complicated barrel cam, and even more for a three-dimensional cam. A less expensive (nonservo) motor can be used to drive the camshaft.

Reliability

Once a properly designed cam is properly manufactured, it can last a very long time. If it is properly lubricated, one should expect many millions of cycles from the cam and follower before it fails. Modern automobile engines are capable of going 200 000 miles without camshaft replacement. Conservatively assuming an average speed of 35 mph and an average engine speed of 1500 rpm, this amounts to about a quarter-billion camshaft revolutions without failure. Cams of the sort used in production machinery do not lead such a happy life since they are not bathed in filtered oil, but are often run dry, or with minimal lubrication, in less than pristine environments. Nevertheless, some production machine cams last several years on machines that accumulate as many as 100 million cycles per year. Servomotors and controllers have high reliability as well. Electronic components can fail and bearings can wear out, but these systems have long lives if properly sized.

Complexity

Cams and servo systems both require engineering-level personnel to be correctly designed. Specialized training in cam design or servo programming and tuning is needed for these tasks. The mathematical requirements for their motion functions are essentially the same for both systems. Dynamics is dynamics whether achieved by servo or cam. F always equals mA . The ratio of load inertia to servomotor inertia must be kept in proper range by the designer to achieve the desired dynamics. A cam does not have this requirement. The desired servo motion program can only be achieved by proper tuning of the system controller. If no changes are made to the driven system, the tuning can remain untouched for the life of the machine. Once a properly designed cam is manufactured, it needs no further tuning. If servo systems require some reprogramming during their lifetimes, it is typically done by highly trained personnel. Cams can be serviced by mechanics and technicians.

Robustness

Cam-driven machines maintain synchrony and phasing between mechanisms mechanically. If the power fails, the machine comes to a stop with no change in phasing. If power is lost to a servo-driven system, all control over relative position is lost and crashes can

occur with expensive consequences. Servo controller programs also have been known to lose their phasing even with no power loss and then require a machine stop to re-home the drives, meaning to get them all back to their zero position.

Packaging

Cam-follower systems can be quite compact (think of an overhead camshaft valve train) or large if the motion must be transferred over some distance, as in many assembly machines with camshafts below and tooling above the base plate. In some cases, a small servomotor mounted at the tooling can provide the needed motions without the complexity of the linkage system. Linear servos lend themselves well to this situation if the desired output motion is rectilinear translation, as they can be mounted to act directly on the end effector. They can achieve large displacements also. The resultant reduction of mass and increased stiffness of the follower train can have significant dynamic advantages.

Load Capacity

Cams have inherently high mechanical advantage. This makes them very suitable for applications that require the generation of high forces or torques or ones that have large inertia loads. One example is the compaction of the contents in a dry-cell battery. The required forces are thousands of pounds, and cams are typically used to drive a compacting ram into the battery to reduce its volume. Another is the insertion/bonding of lid liners in food jars which requires around 7000 lb force, also done with cam mechanisms.

A servomotor driving the input crank to a linkage or other mechanism through a gearbox has the gearbox ratio as a constant mechanical advantage. The linkage, of course, introduces its own additional, variable mechanical advantage. The same linkage, driven by a cam acting on a roller attached to the input crank, substitutes a variable mechanical advantage for that of the servo gearbox's constant one. A linear servomotor acting directly on the end effector has no mechanical advantage. Though these devices can generate large forces, a cam with its infinite mechanical advantage will still generate much more force.

As an example, consider the same slider-crank linkage to be driven through the same crank angle by either a cam acting on a roller placed at the joint between crank and connecting rod or a servomotor driving the crank directly. Each can provide the identical angular displacement, velocity, and acceleration profiles to the crank. The mass of the linkage is the same in both cases. Let the motion profile be a 3-4-5-6 polynomial rise-fall-dwell function applied to the crank. Calculate the torque required by the motor driving the cam and the motor directly driving the crank and you will find that the servomotor directly driving the crank must apply a torque about 20 times that required on the camshaft. Now, a 20:1 reduction gearbox will even things out, after a fashion. But, if the purpose of the mechanism is to provide large external forces at some point in the stroke, the cam's variable (and potentially infinite) mechanical advantage may be preferable to the gearbox's constant mechanical advantage.

When dealing with high speeds and high loads, the motor's so-called velocity-torque envelope can become a limitation. The product of angular velocity and torque is power, and any motor has a particular angular velocity-torque curve as shown in Figures 2-40 through 2-42. When a servomotor's size is increased to obtain the required torque for a high-load application, its rotor's polar moment of inertia also increases as the square of

its radius. The result is a disproportionate amount of torque being required to accelerate the rotor and thus not being available to drive the load. This results in an oversized motor compared to one driving a cam that runs at constant speed and amplifies the force or torque through its variable mechanical advantage. A known example is an application that uses 47% of the servomotor's torque capacity just to accelerate the motor rotor each cycle.

16.7 REFERENCES

- 1 Norton, R. L. (2009). *Cam Design and Manufacturing Handbook* 2ed. Industrial Press Inc.: New York, pp. 417-420.
- 2 Ibid, Chapter 5.

16.8 BIBLIOGRAPHY

Dote, Y., and S. Kinoshita (1990). *Brushless Servomotors*. Clarendon Press: Oxford.

Humphrey, W. M. (1973). *Introduction to Servomechanism System Design*. Prentice-Hall: Englewood Cliffs, NJ.

Younkin, G. W.. (2003). *Industrial Servo Control Systems*. Marcel Dekker: New York.

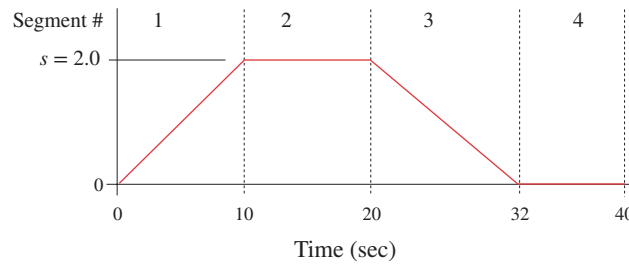
16.9 PROBLEMS

Program DYNACAM may be used to solve these problems where applicable. Where units are unspecified, work in any consistent units system you wish.

- 16-1 Design a double-dwell motion to translate a slider from 0 to 75 mm in 60° with modified sine acceleration, dwell for 120°, fall 75 mm in 30° with cycloidal motion, and dwell for the remainder. Calculate and plot the displacement function of the slider using DYNACAM and modify that function to drive the crank of a nonoffset crank-slider mechanism with dimensions: crank = 100 mm, coupler = 300 mm such that at maximum stroke of the slider, the crank is at zero degrees. The slider must follow the specified motion. Plot the resulting crank input function.
- 16-2 Repeat Problem 16-1 for a slider function for a 65-mm rise with 3-4-5 polynomial motion over 80°, dwell for 100°, fall over 40° with 3-4-5 polynomial motion, and dwell for the remainder.
- 16-3 Design a single-dwell polynomial motion to translate a slider from 0 to 75 mm and return to 0 in 120° and dwell for the remainder. Total cycle time is 6 sec. Calculate and plot the displacement function of the slider using DYNACAM and modify that function to drive the crank of a non-offset crank-slider mechanism with the dimensions crank = 90 mm, coupler = 270 mm such that at maximum stroke of the slider, the crank is at 180°. The slider must follow the specified motion. Plot the resulting crank input function.
- 16-4 Using the constant velocity motion program of Example 16-1 applied to the slider of a nonoffset fourbar crank-slider linkage with a constant velocity stroke of 150 mm at 40 mm/s, calculate and plot the s , v , a , j functions for this motion applied directly to the slider using program DYNACAM. Total cycle time is 6 sec. Calculate the modified program needed to drive the crank of the linkage to obtain the specified motion at the slider. Crank = 200 mm, coupler = 600 mm, crank start angle = 110°.

TABLE P16-0
Topic/Problem Matrix

16.3 Cam-Driven Linkages
16-1 to 16-5
16.4 Servo-Driven Linkages
16-6, 16-7

**FIGURE P16-1**

Timing diagram for Problems 16-6 and 16-7

- 16-5 Repeat Problem 16-4 for slider stroke of 5.5 in at 0.75 in/sec, a crank length of 7.5 in, a coupler length of 18 in, and a crank start angle of 120° . Total cycle time is 12 sec.
- 16-6 A nonoffset crank-slider with crank and coupler lengths of 3.25 in and 10.875 in, respectively, is to be driven by a servo through a gear reducer. The required slider timing diagram is shown in Figure P16-1. The slider moves through the distance s in. Choose a set of servo functions from Table 16-2 to drive the crank with the lowest resulting vibration and calculate the resulting crank displacement for one cycle of 40-sec duration. Based on this crank input, calculate the resulting slider motion and compare the two by plotting their normalized displacements. The crank angle at $t = 0$ is 70° , and the crank turns clockwise to accomplish the first motion of the slider.
- 16-7 Repeat Problem 16-6 using the timing diagram of Figure P16-1 but with a total stroke of 75 mm. The crank length is 110 mm, the coupler length is 275 mm, and the initial crank angle is 90° .